

Practice Problem Set 1

(1) Sketch, draw, or otherwise determine the subsets of $\mathbb{C}$ defined by the following relations:

- (a) $|z + i| \geq |z - i|$;
- (b) $0 < |z| < 1$;
- (c) $\frac{1}{2} \leq |z| < 1$;
- (d) $\operatorname{Re}(1/z) < 1$.

Which of the sets described above are closed? Open? Neither open nor closed?

(2) **(Polar representation of complex numbers, arguments, powers and roots)** Consider the mapping $\Phi : (0, \infty) \times \mathbb{R} \rightarrow \mathbb{C} \setminus \{0\}$ defined as follows:

$$\Phi(r, \theta) := (r \cos \theta) + i(r \sin \theta),$$

where, as usual, we think of the first input of Φ as “the radius” or “the radial input” and denote it by r and of the second input of Φ as “the angle” or “the angular input” and denote it by θ . Let z be an arbitrary *non-zero* complex number and consider the number $z/|z|$. It is a complex number of unit modulus, which means it can be written as $x + iy$, where $x, y \in \mathbb{R}$ and $x^2 + y^2 = 1$. It is a fundamental fact that in this situation there exists a unique $\theta = \theta_z \in [0, 2\pi)$ such that $\cos \theta_z = x$ and $\sin \theta_z = y$. Consequently, we can write

$$\frac{z}{|z|} = x + iy = \cos \theta_z + i \sin \theta_z,$$

so that

$$z = (|z| \cos \theta_z) + i(|z| \sin \theta_z) = \Phi(|z|, \theta_z).$$

The above discussion shows that Φ is onto (a.k.a. surjective). However, Φ is highly non-injective. Namely, if $z \in \mathbb{C} \setminus \{0\}$ and θ_z is as above, then

$$\forall k \in \mathbb{Z}, z = \Phi(|z|, \theta_z + 2k\pi).$$

(a) Show that, if $(r, \theta) \in (0, \infty) \times \mathbb{R}$ and $\Phi(r, \theta) = z$, then $r = |z|$ and $\exists k \in \mathbb{Z}$ such that $\theta = \theta_z + 2k\pi$. From the above it follows that

$$\forall z \in \mathbb{C} \setminus \{0\}, \Phi^{-1}\{z\} = \{(|z|, \theta_z + 2k\pi) \mid k \in \mathbb{Z}\}.$$

The map Φ has several nice properties. Two of the most interesting are the following:

(b) Show that, if $(r_1, \theta_1), (r_2, \theta_2) \in (0, \infty) \times \mathbb{R}$, then

$$\Phi(r_1, \theta_1)\Phi(r_2, \theta_2) = \Phi(r_1 r_2, \theta_1 + \theta_2).$$

Note that, in the above equation, juxtaposition on the LHS denotes multiplication of complex numbers. By induction, it follows that, for an arbitrary positive integer n , if $(r_1, \theta_1), (r_2, \theta_2), \dots, (r_n, \theta_n) \in (0, \infty) \times \mathbb{R}$,

$$\Phi(r_1, \theta_1)\Phi(r_2, \theta_2) \dots \Phi(r_n, \theta_n) = \Phi(r_1 r_2 \dots r_n, \theta_1 + \theta_2 + \dots + \theta_n).$$

(c) Show that, if $(r, \theta) \in (0, \infty) \times \mathbb{R}$ and n is an arbitrary integer, not necessarily positive, then

$$\Phi(r, \theta)^n = \Phi(r^n, n\theta).$$

Using the above discussion, prove the following important fact:

(d) For every $z \in \mathbb{C} \setminus \{0\}$ and every $n \in \mathbb{Z}_+$, there exist exactly n distinct complex numbers w such that $w^n = z$. What are they in terms of z ?

The non-injectivity of Φ is somewhat dissatisfying. One may forcefully try to make the map injective. Indeed, what we have discussed above shows that Φ is injective *when restricted to the sub-domain* $(0, \infty) \times [0, 2\pi)$ of its original domain of definition, $(0, \infty) \times \mathbb{R}$. In mathematical terms,

$$\Phi|_{(0, \infty) \times [0, 2\pi)} \text{ is injective.}$$

Our goal in doing this, is, of course, to recover the “argument of complex numbers” with which you have probably been familiar since school. However, we will show that this is, in a precise sense, impossible. Let Ψ denote the inverse of the restricted map above, i.e.,

$$\Psi := (\Phi|_{(0, \infty) \times [0, 2\pi)})^{-1};$$

then Ψ is a map from $\mathbb{C} \setminus \{0\}$ to $(0, \infty) \times [0, 2\pi)$. Hence we can write $\Psi = (\Psi_1, \Psi_2)$, where $\Psi_1 : \mathbb{C} \setminus \{0\} \rightarrow (0, \infty)$ and $\Psi_2 : \mathbb{C} \setminus \{0\} \rightarrow [0, 2\pi)$. If the “argument of complex numbers” is to mean anything at all, it must mean Ψ_2 . The problem is that Ψ_2 is not a nice mapping at all. Here is a somewhat challenging exercise:

(e) Show that $\Psi_2 : \mathbb{C} \setminus \{0\} \rightarrow [0, 2\pi)$ is not continuous.

Furthermore, the most important property that the “argument” is supposed to satisfy, when expressed in terms of Ψ_2 is

$$\forall z_1, z_2 \in \mathbb{C} \setminus \{0\}, \Psi_2(z_1 z_2) = \Psi_2(z_1) + \Psi_2(z_2).$$

(The colour is meant to warn you that the equation does *not* hold.) Indeed, it is easy to see that the equation does not hold.

(f) Show that there exist $z_1, z_2 \in \mathbb{C} \setminus \{0\}$ such that

$$\Psi_2(z_1 z_2) \neq \Psi_2(z_1) + \Psi_2(z_2).$$

There is a way to (partially) rectify the above unsatisfactory situation, but it involves making a sacrifice. That is, there is a way to make the map Ψ continuous. There is no *natural* way of doing this, but there is a standard way, which is as follows. Restrict Φ , originally defined on $(0, \infty) \times \mathbb{R}$, to $(0, \infty) \times (-\pi, \pi)$.

(g) Let $V := (0, \infty) \times (-\pi, \pi)$ and let $U := \mathbb{C} \setminus \{x + iy \mid x \leq 0, y = 0\}$. Show that $\Phi(V) = U$.

(h) Show that $\Phi|_V$ is injective.

Redefine Ψ to be $(\Phi|_V)^{-1} : U \rightarrow V$.

(i) Show that Ψ is continuous.

One may define $\arg : U \rightarrow (-\pi, \pi)$ by $\arg(z) := \Psi_2(z)$; this is called **the standard (branch of the) argument**. However, one *still* does not have $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$ for all $z_1, z_2 \in U$, even when $z_1 z_2 \in U$:

(j) Show that there exist $z_1, z_2 \in U$ such that $z_1 z_2 \in U$ and such that $\arg(z_1 z_2) \neq \arg(z_1) + \arg(z_2)$.

(3) Prove that, if $f : \mathbb{C} \rightarrow \mathbb{R}$ is a continuous function, then, for every $a \in \mathbb{R}$, both

$$f^{-1}((-\infty, a)) := \{z \in \mathbb{C} \mid f(z) < a\} \quad \text{and}$$

$$f^{-1}((a, \infty)) := \{z \in \mathbb{C} \mid f(z) > a\}$$

are open subsets of $\mathbb{C}$.

(4) In each of the cases below, determine whether f is continuous at 0, where $f : \mathbb{C} \rightarrow \mathbb{C}$ is the function such that $f(0) = 0$ and $f(z)$, for $z \neq 0$, is given by

$$(a) \frac{\operatorname{Re}(z^2)}{|z|};$$

$$(b) \frac{\operatorname{Im}(z^2)}{|z|};$$

$$(c) \frac{\operatorname{Re}(z^2)}{|z|};$$

$$(d) \frac{\operatorname{Im}(z)^2}{|z|};$$

$$(e) \frac{\operatorname{Re}(z^2)}{|z|^2};$$

$$(f) \frac{\operatorname{Im}(z)^2}{|z|^2}.$$

(5) Determine the following limits (i.e., show that they exist and find out what they are):

$$(a) \lim_{z \rightarrow 3i} \frac{z^2 + 9}{z - 3i}; \quad (b) \lim_{z \rightarrow 3i} \frac{z^3 + 27i}{z^2 + 9}.$$

- (6) Let n be an arbitrary positive integer. Show that the following limit exists and find it:

$$\lim_{z \rightarrow 1} \frac{1 - z^n}{z^2 + 5z - 6}.$$

- (7) Show that the function $f : \mathbb{C} \rightarrow \mathbb{C}$ defined by $f(z) := z^2 \bar{z}$ is $\mathbb{C}$ -differentiable at just one point and find out what this point is.
- (8) For a, b, c, d fixed but arbitrary elements of $\mathbb{R}$, define $f : \mathbb{C} \rightarrow \mathbb{C}$ by

$$\forall z \in \mathbb{C}, f(z) := (a\operatorname{Re}(z) + b\operatorname{Im}(z)) + i(c\operatorname{Re}(z) + d\operatorname{Im}(z)).$$

Show that f is $\mathbb{C}$ -differentiable at 0 iff it is $\mathbb{C}$ -differentiable on $\mathbb{C}$ iff $a = d$ and $b = -c$. Can you find a more natural way to write f ?

- (9) Define $f : \mathbb{C} \rightarrow \mathbb{C}$ as follows:

$$\forall z \in \mathbb{C}, f(z) := |z|^2.$$

Show that f is $\mathbb{C}$ -differentiable only at 0.

- (10) Define $f : \mathbb{C} \rightarrow \mathbb{C}$ as follows:

$$\forall z \in \mathbb{C}, f(z) := \operatorname{Re}(z)\operatorname{Im}(z) + i\operatorname{Im}(z).$$

Show that f is $\mathbb{C}$ -differentiable at exactly one point of $\mathbb{C}$, and find this point.

- (11) **(A definition of the exponential function)** We want to extend the exponential function $\exp$, defined on $\mathbb{R}$, to $\mathbb{C}$. Note that the most important feature of $\exp$ is its behaviour under addition of its inputs:

$$\forall x, y \in \mathbb{R}, \exp(x + y) = \exp(x)\exp(y);$$

note also that every complex number z can be written uniquely as $x + iy$ with $x, y \in \mathbb{R}$. Therefore, if we want the additive property above to be satisfied, we see that, in order to extend $\exp$ to $\mathbb{C}$, it is sufficient to define $\exp(iy)$ for $y \in \mathbb{R}$. Also check that, if we somehow manage to define $\exp$ on $\mathbb{C}$ so that it satisfies the additive property above and agrees with $\exp$ on $\mathbb{R}$, then we must have $\exp(0) = 1$. Let us assume that $\exp(iy) = u(y) + iv(y)$ with u, v real-valued functions and try to determine what u, v must be. First conclude, using the previous remark, that one must have $u(0) = 1$ and $v(0) = 0$. Let us also assume, for the sake of simplicity, that u, v are differentiable (this is a mild assumption). Then our tentative definition of $\exp$ becomes $\exp(x + iy) = \exp(x)(u(y) + iv(y))$. Show that, with this definition, $\exp$ is $\mathbb{C}$ -differentiable if and only if $u(y) = v'(y)$ and $u'(y) = -v(y)$. Surely we want $\exp$ to be $\mathbb{C}$ -differentiable (it will, in fact, turn out to be infinitely differentiable, but we'll see that later.) Therefore, the equations above should suggest something! Show that, if we define $\exp(iy) := \cos(y) + i\sin(y)$ for $y \in \mathbb{R}$ arbitrary, then $\exp$, defined on $\mathbb{C}$ as follows:

$$\forall z \in \mathbb{C}, f(z) := \exp(\operatorname{Re}(z))(\cos(\operatorname{Im}(z)) + i\sin(\operatorname{Im}(z)))$$

is $\mathbb{C}$ -differentiable everywhere. (It is a theorem that if (u, v) is a pair of differentiable functions on $\mathbb{R}$ such that $u(0) = 1$, $v(0) = 0$, $u' = -v$ and $v' = u$ then $u = \cos$ and $v = \sin$. This shows that the definition

$$\forall y \in \mathbb{R}, \exp(iy) = \cos(y) + i\sin(y)$$

is not arbitrary but is the only reasonable definition.)

- (12) With the complex exponential defined as above, define $f : \mathbb{C} \rightarrow \mathbb{C}$ as follows:

$$\forall z \in \mathbb{C}, f(z) := \exp(\operatorname{Im}(z))\exp(i\operatorname{Re}(z)).$$

Show that f is $\mathbb{C}$ -differentiable nowhere.